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Introduction and rationale

Many people want to separate their waste correctly, but they do not recognize the material or know the rules that apply in their location. This uncertainty leads to incorrect disposal and reduces the amount of material that reaches recycling systems.

Wasted will let users photograph an item and request a classification. Google Cloud Vision will identify objects, text, and visual labels.³ Our system will compare those results with local rules, then display the material, category, and recommended preparation.

Wasted addresses SDG 12: Responsible Consumption and Production by focusing on correct waste separation and recycling.¹ It also addresses part of SDG 11: Sustainable Cities and Communities because it improves waste management in cities and encourages community participation.²

This problem occurs every day, so the team can test the proposal with real objects and common disposal decisions. The project combines interface design, web development, API integration, data handling, and privacy within a scope that fits the semester.



Problem and proposed idea

People need clear information at the moment they discard an item. Most guides assume that users already know the material, while collection and recycling rules vary by location.

We will test whether visual recognition reduces this uncertainty without hiding its errors. We will design a prototype that explains each result, reports its confidence level, and accepts corrections from the user.

The user will complete four steps:

1. The user takes a photograph or selects an existing image.
2. Wasted sends the image to the API and analyzes the returned labels.
3. Wasted displays the category, explanation, and preparation instructions.
4. The user confirms the result or selects another option.

Wasted will process the photograph without storing it by default. With consent, the system will store the labels, confidence score, selected location, and voluntary correction. Google processes images from online requests in memory and does not retain them on disk.⁴

The first version will cover one location and a limited set of common waste categories. It will include text search, and the interface will preserve the same sequence on mobile and desktop devices.



Professor's observations

The review of this proposal raised four observations. Each one is answered here with the decision we took and the way we will implement it.

1. "I wonder how would this keep the user engagement."

Wasted will keep a daily streak: one classified item keeps the day alive. Long streaks unlock rewards from sponsors, an environmental brand or a public program. A consumer goods company could offer a coupon for an ecological dish soap after twenty consecutive days. Every reward has to be an ecological product or reduce the carbon footprint.

2. "Once you have enough information from a user, you may be able to suggest consumption improvements by visualizing their waste patterns and offering alternatives."

We take this as the growth path, in a deliberately small first version. With consent, Wasted stores each classification and, once there are enough entries, shows a chart of categories over time and a short list of ecological alternatives for the items that repeat the most. We will not predict consumption: that needs models and a data volume that do not fit this semester.

3. "One limitation for this app would be the location rules, that could be tricky because maybe there is not easily accessible information about that, you may need to create such DB."

The database already exists and we verified it. SEMADET publishes its register of authorised sites as GeoJSON through the Jalisco map viewer: 524 collection and recycling points, 420 of them in the metropolitan area.⁵ Guadalajara and Zapopan publish their citizen containers the same way.^{6,7} Names, telephones and accepted materials come from the list of 562 companies that SEMADET published in September 2026.⁸ With permission, Wasted reads the user location, matches the material with the nearest point that accepts it, and opens the route in Google Maps or Waze.

4. "Make sure that your interaction with the CV API works properly for your purposes, I don't know if there is already a data set for this purposes."

This observation changed our approach: instead of reading generic labels, the photograph goes to a vision language model that must answer in a fixed JSON schema with the object, its material, its category and the preparation it needs. Our server validates that answer, looks for the disposal point and renders the result, so no training dataset is required. We will keep our own set of test photographs to measure how often the answer is correct and the schema respected.



Benefits and conclusion

Wasted will reduce uncertainty, explain separation rules, and identify the types of waste that cause the most errors. It will convert an image recognition result into a specific action based on the user's location.

We will test the prototype with a controlled image set and short user sessions. We will measure classification accuracy, response time, corrections, and perceived usefulness. If the confidence level remains low, Wasted will request another photograph or present alternative classifications.

Wasted will combine computer vision, local rules, and human decisions. The project will succeed if it responds quickly when it recognizes an item and reports uncertainty when the available evidence does not support a reliable classification.

1. United Nations, Goal 12: Responsible Consumption and Production. sdgs.un.org/goals/goal12.

2. United Nations, Goal 11: Sustainable Cities and Communities. sdgs.un.org/goals/goal11.

3. Google Cloud, Detect Labels — Cloud Vision API. docs.cloud.google.com/vision/docs/labels.

4. Google Cloud, Data Usage FAQ — Cloud Vision API. docs.cloud.google.com/vision/docs/data-usage.

5. SEMADET, Gobierno de Jalisco, Mapa Jalisco, gestión integral de residuos. mapa.jalisco.gob.mx (WFS geoserver/geoserver/vjallows).

6. Gobierno de Guadalajara, Servidor de mapas, medio ambiente. mapa.guadalajara.gob.mx (WMS geoserver/medio_ambiente).

7. Gobierno de Zapopan, Servidor de mapas, geomatica. mapa.zapopan.gob.mx (WFS geoserver/geomatica/ows).

8. SEMADET, Residuos de manejo especial, listado de empresas autorizadas. semadet.jalisco.gob.mx/medio-ambiente/residuos/residuos-de-manejo-especial.

Wasted
throw it right